Contextual Speech Synthesis for Study Materials

Abstract. Is it possible to read academic documents aloud naturally? We define natural narration as speech that a listener cannot distinguish from a human reader.

1. Introduction

Reading long documents aloud is demanding. The method was evaluated on three independent datasets. Results indicate a consistent improvement over the baseline. However, the effect size varies considerably between conditions. We therefore report confidence intervals for every estimate. This observation motivates the analysis in the following section. Several limitations must be acknowledged before drawing conclusions. Consider the equation $x^2 + y^2 = z^2$, which a narrator must speak as words rather than symbols.

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2. Method

We define the error rate as the fraction of mispronounced words. This observation motivates the analysis in the following section. Several limitations must be acknowledged before drawing conclusions. In particular, the sample was drawn from a single institution. Future work should replicate these findings in other settings. The method was evaluated on three independent datasets. Results indicate a consistent improvement over the baseline. However, the effect size varies considerably between conditions. We therefore report confidence intervals for every estimate. This observation motivates the analysis in the following section. Several limitations must be acknowledged before drawing conclusions.

1. Extract text in reading order.
2. Normalize numbers, units and symbols.
3. Synthesize speech with word timing.

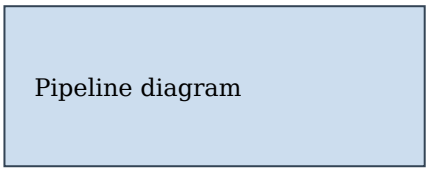


Figure 1: Overview of the narration pipeline.

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3. Results

The mean error decreased from 4.1% to 1.7% ($p < 0.01$). We therefore report confidence intervals for every estimate. This observation motivates the analysis in the following section. Several limitations must be acknowledged before drawing conclusions. In particular, the sample was drawn from a single institution. Future work should replicate these findings in other settings. The method was evaluated on three independent datasets. Results indicate a consistent improvement over the baseline. However, the effect size varies considerably between conditions. We therefore report confidence intervals for every estimate. This observation motivates the analysis in the following section. Several limitations must be acknowledged before drawing conclusions. In particular, the sample was drawn from a single institution. Future work should replicate these findings in other settings. The method was evaluated on three independent datasets. Results indicate a consistent improvement over the baseline. However, the effect size varies considerably between conditions.

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4. Conclusion

In conclusion, structure-aware preparation improves narration quality. In particular, the sample was drawn from a single institution. Future work

should replicate these findings in other settings. The method was evaluated on three independent datasets. Results indicate a consistent improvement over the baseline. However, the effect size varies considerably between conditions. We therefore report confidence intervals for every estimate. This observation motivates the analysis in the following section. Several limitations must be acknowledged before drawing conclusions. In particular, the sample was drawn from a single institution. Future work should replicate these findings in other settings. The method was evaluated on three independent datasets. Results indicate a consistent improvement over the baseline.

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1 Footnote: all experiments were run on a standard laptop.